Predicting Coral Bleaching Using Linear and Polynomial Regression

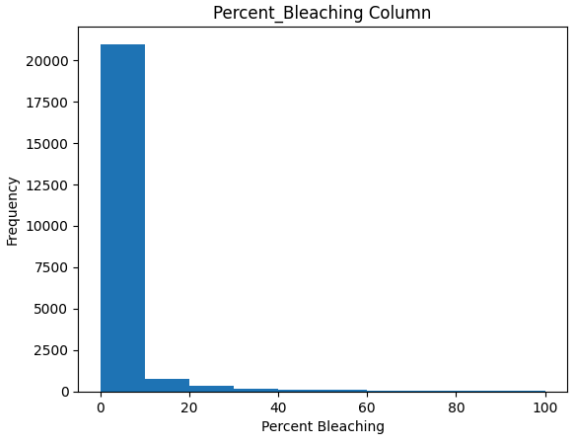
Introduction & Dataset

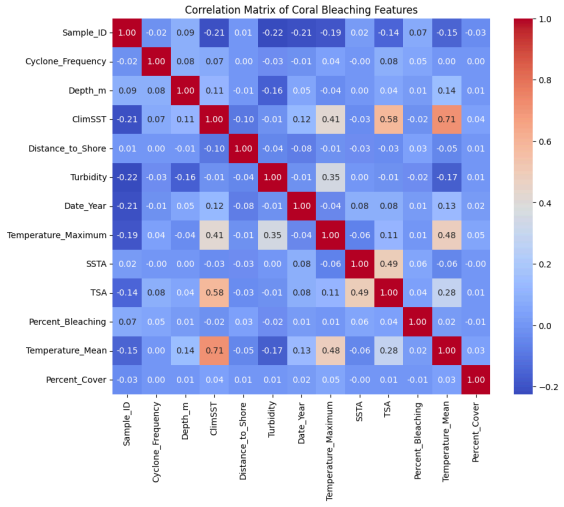
Coral bleaching occurs when environmental stress causes corals to lose the symbiotic algae that provide much of their energy and color. Although bleached corals are not necessarily dead, prolonged bleaching can lead to starvation and eventually death. Rising ocean temperatures are the leading cause of large-scale coral bleaching, but other factors such as pollution, sunlight, and water quality can also affect coral health.

Climate change and environmental conservation are topics I've become increasingly interested in. While looking for datasets, I came across coral bleaching, something I previously knew little about. Coral reefs support marine biodiversity, so understanding the conditions that contribute to bleaching is important.

For this project, I used the Coral Reef Global Bleaching dataset, originally published through BCO-DMO but I got it from Kaggle. I focused on observations measuring bleaching at the population level.

The goal was to predict Percent_Bleaching using environmental and geographic variables such as temperature, depth, turbidity, cyclone frequency, and ocean region.





Data Exploration and Feature Engineering

The distribution of bleaching percentages was heavily right-skewed, with approximately 93% of observations having less than 10% bleaching. Since severe bleaching was relatively uncommon, I expected the regression models might struggle to predict higher bleaching percentages.

While examining feature histograms, I also discovered 3,646 observations with a ClimSST value of 262.15 K (−11°C), an unrealistic temperature for coral reef environments. The correlation matrix showed that ClimSST had a correlation of 0.71 with Temperature_Mean. Because of these suspicious values and the missing data, I removed ClimSST while keeping Temperature_Mean.

I also removed unnecessary identifiers and checked for duplicate observations. For preprocessing, I used KNN imputation for missing numerical values, standardized numerical features, one-hot encoded ocean regions, and ordinal encoded reef exposure. I used an 80/20 train/test split, keeping observations from the same sampling event together to prevent data leakage.

Another limitation was geographic imbalance: approximately 67.7% of observations came from the Pacific Ocean, which could affect how well the model generalizes to other regions.

Clustering Analysis

I used K-means clustering to investigate whether coral reefs formed distinct groups based on environmental conditions, without using Percent_Bleaching.

The elbow method did not reveal a clear optimal number of clusters. I then compared silhouette scores for K = 2 through K = 8. The highest score was only 0.127 at K = 2, suggesting weakly separated clusters.

The two clusters also had similar average bleaching percentages of 2.62% and 2.28%, with both having a median of 0%. This suggests that the environmental features did not form distinct groups associated with substantially different bleaching levels.

Regression Models and Evaluation

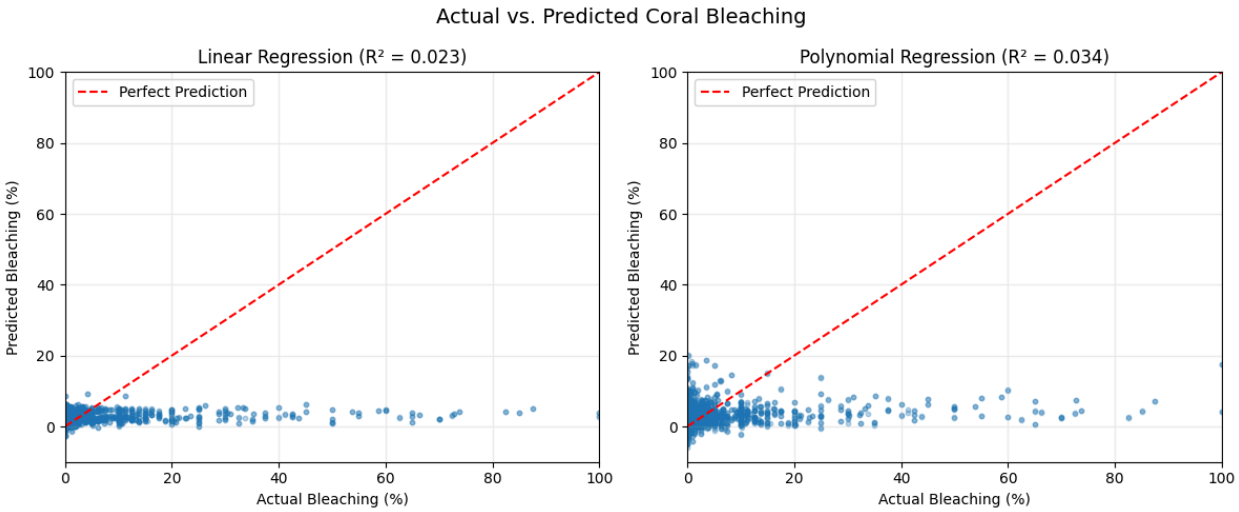
I trained linear and degree-2 polynomial regression models to predict bleaching percentages. I evaluated both using R^2 and mean absolute error (MAE).

Model	R^2	Overall MAE	MAE (≥20% Bleaching)
Linear Regression	0.023	3.86	37.26
Polynomial Regression	0.034	3.95	36.13

MAE values are measured in percentage points.

Polynomial regression explained slightly more variation and performed somewhat better for severe bleaching, while linear regression had a lower overall MAE. However, neither model was particularly effective.

Although the overall errors appeared relatively small, the models performed poorly when actual bleaching reached at least 20%. This suggests that the large number of observations with little or no bleaching may have made the overall MAE misleading.



Conclusion

Both regression models struggled to accurately predict coral bleaching, explaining less than 4% of the variation in bleaching percentages. Although polynomial regression showed a small improvement, neither model was effective at predicting severe bleaching.

These results suggest that the environmental variables available in this dataset may not be enough to accurately predict bleaching using the models tested. Other factors, such as coral species, historical heat exposure, symbiotic algae, and local environmental conditions, could also influence how corals respond to stress.

This project showed me that having a large dataset does not necessarily mean a machine learning model will make accurate predictions. It also raised a larger question: What additional biological and environmental factors would help us better understand and predict coral bleaching?

[Github Page]
Jupyter Notebook